Stochastic Processes
Spring 2025 April 29, 2025

Homework #5: Continuous Time Markov Chains

Problem 1. Consider three machines that are maintained by a single repairman. Machine i functions for an exponential time with rate μ_i before breaking down, $i = 1, 2, 3$. The repair times (for either machine) are exponential with rate μ. Can we analyze this as a birth and death process? If so, what are the parameters? If not, how can we analyze it? (Note that the repair man can only fix one machine at the same time.)

Problem 2 (Chapter 6, Exercise 17). Each time a machine is repaired it remains up for an exponentially distributed time with rate λ. It then fails, and its failure is either of two types. If it is a type 1 failure, then the time to repair the machine is exponential with rate μ_1; if it is a type 2 failure, then the repair time is exponential with rate μ_2. Each failure is, independently of the time it took the machine to fail, a type 1 failure with probability p and a type 2 failure with probability $1 - p$. What proportion of time is the machine down due to a type 1 failure? What proportion of time is it down due to a type 2 failure? What proportion of time is it up?

Problem 3 (Chapter 6, Exercise 22). Customers arrive at a single-server queue in accordance with a Poisson process having rate λ. However, an arrival that finds n customers already in the system will only join the system with probability $1/(n + 1)$. That is, with probability $n/(n + 1)$ such an arrival will not join the system. Show that the limiting distribution of the number of customers in the system is Poisson with mean λ/μ. Assume that the service distribution is exponential with rate μ.

Problem 4 (Chapter 6, Exercise 23). A job shop consists of three machines and two repairmen. The amount of time a machine works before breaking down is exponentially distributed with mean 10. If the amount of time it takes a single repairman to fix a machine is exponentially distributed with mean 8, then

(a) what is the average number of machines not in use?

(b) what proportion of time are both repairmen busy?

Problem 5 (Chapter 6, Exercise 38). Consider an n server system where the service times of server i are exponentially distributed with rate μ_i, $i = 1, ..., n$. Suppose customers arrive in accordance with a Poisson process with rate λ, and that an arrival who finds all servers busy does not enter but goes elsewhere. Suppose that an arriving customer who finds at least one idle server is served by a randomly chosen one of that group; that is, an arrival finding k idle servers is equally likely to be served by any of these k servers.

(a) Define states so as to analyze the preceding as a continuous-time Markov chain.

(b) Show that this chain is time reversible.

(c) Find the limiting probabilities.

Problem 6. Consider an M/M/∞ queue where customers arrive according to a Poisson process with rate $\lambda = 5$ per minute. Each customer requires an exponentially distributed service time with mean $1/\mu = 1/3$ minutes.

(a) What is the probability that exactly $n = \mathbf{4}$ customers are in the system at a random point in time?

(b) What is the expected number of customers in the system?

(c) What is the variance of the number of customers in the system?